The SHA-256 + CIE Protocol — Operational Guide

> *Full paper:* *docs/Certifying_a_Work_of_Art_v12_EN.docx*

The 5 Layers at a Glance

```
Layer 1  →  PDF/A Certificate of Authenticity
Layer 2  →  SHA-256 hash of the PDF
Layer 3  →  SHA Document (second PDF/A containing the hash)
Layer 4  →  CIE digital signature on both PDFs  ← mandatory
Layer 5  →  Certified email / PEC               ← mandatory
```

Order is binding. Each layer executes after the previous one.

Layer 1 — Certificate of Authenticity

Compose in Word, LibreOffice, or Canva. Export as **PDF/A-2b** (not standard PDF).

Required fields: - Full name (as on identity document) - Title, year, technique and materials (precise: not “print” but “inkjet on Hahnemühle 310g”) - Exact format in cm - Copy number and total edition (e.g. 3/10 or A.P.) - SHA-256 hash of the source image file (see below) - SHA-256 hash of the print photograph (see Part 8.0 of the paper) - Edition clause (see paper for exact wording)

Layer 2 — Calculate the Hash

Use `sha_generator_en.html` — offline, no installation.

Or via command line:

```
# Windows
certutil -hashfile certificate.pdf SHA256

# Mac / Linux
shasum -a 256 certificate.pdf

# SHA-512 (recommended as secondary)
shasum -a 512 certificate.pdf

# BLAKE2b (optional, additive only — never replace SHA-256)
python3 -c "import hashlib;
print(hashlib.blake2b(open('certificate.pdf','rb').read(),
digest_size=32).hexdigest())"
```

Algorithm hierarchy: | Algorithm | Status | Characters | |———–|———–|———–| | SHA-256 | **Mandatory** (ISO, NIST, eIDAS) | 64 | | SHA-512 | Recommended secondary | 128 | | BLAKE2b-256 | Optional additional layer | 64 | | MD5, SHA-1 | **Never use** — cryptographically compromised | — |

Layer 3 — SHA Document

A second PDF/A containing: - File name and size - Date and time of hash calculation - Complete SHA-256 string (64 characters) - Complete SHA-512 string (128 characters, if calculated) - Author’s personal details

Use `sha_generator_en.html` → it generates the pre-formatted text automatically.

Layer 4 — CIE Digital Signature

Sign **both** PDFs (certificate + SHA document) with your eID card.

Tools: - **Firma4NG** (PC — Windows/Mac/Linux): free, from agid.gov.it - **CieSign** (iOS/Android): official Ministry of the Interior app

Select **PAdES** format (embedded in the PDF). Not CAdES (.p7m separate file).

Legal note — AES vs QES: - CIE signature = **Advanced Electronic Signature (AES/FEA)** - In disputes: burden of proof falls on the party presenting the document - For works > €5,000 or high-risk contexts: consider a **Qualified Electronic Signature (QES)** via CNS token

After signing: export the `.p7s` validation report and keep it with the signed PDFs.

Layer 5 — Certified Email / PEC

Send both signed PDFs to yourself via certified email (PEC in Italy).

- Italian providers: Aruba PEC, Libero Mail PEC, Legalmail (all AgID-accredited)
- EU equivalents: De-Mail (Germany), REM
- The delivery receipt = independent timestamp

Note: PEC is excellent practical redundancy. It is *not* a qualified TSA in the strict eIDAS sense.
For high-value works: add an accredited TSA timestamp (Aruba TSA, InfoCert TSA — a few cents per document).

The Closed Circle (Physical ↔ Digital)

Source file hash (Part 8.1)

Before printing, calculate SHA-256 of your master TIFF/JPEG.
Insert this string in the certificate. This links the digital master to the physical print mathematically.

Print photograph hash (Part 8.0)

In the same CIE signing session: 1. Photograph the physical print (front + back) with a handwritten note in frame: date, copy number, signature 2. Calculate SHA-256 of this photo file 3. Insert in the certificate as “SHA-256 hash — print photograph [N/TOTAL]” 4. Keep the photo file in the work’s archive

Important: This photo is an archival anchor for the artist, not a direct verification tool.
The collector verifies using the PDF hash — not by rephotographing the work.

Transfer of Ownership (Part 13)

When the work changes hands, generate a **Transfer Certificate** — a simple PDF containing:

- SHA-256 hash of the original certificate (cryptographic link)
- Work data: title, copy, year
- Transferor details (name, tax code or passport number)
- Recipient details
- Date of transfer
- Transferor’s declaration

Sign: transferor with CIE, recipient with CIE (or handwritten signature = SES — see paper for legal implications).
Send: both parties send via PEC to each other and to themselves.

Each subsequent sale adds a new Transfer Certificate. The provenance chain grows mathematically, permanently, without any platform.

For non-Italian buyers

- **EU buyers:** eIDAS mutual recognition applies — use national digital signature
- **Non-EU buyers:** handwritten signature + passport number + apostille for high-value works
- Tax code → replace with passport number

Verification (for Collectors)

Use `sha_verifier_en.html` — offline, no installation:

1. Drop the PDF certificate into the page
2. Paste the hash from the SHA Document
3. 🟢 Green = file is authentic and unmodified
4. 🔴 Red = file is different (modified, corrupted, or wrong file)

Or verify in Adobe Acrobat Reader: open the signed PDF → check the signature banner.

Long-Term Preservation

3-2-1 Rule: - 3 copies - 2 different physical media (external drive + USB) - 1 offsite (cloud)

Formats that survive time: - PDF/A (ISO 19005) — self-contained, includes all fonts - SHA-256 / SHA-512 (ISO/IEC 10118-3) — universal, not proprietary - PAdES (ETSI EN 319 102) — independent of any specific software

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Declared Limits

This protocol is honest about what it cannot do:

- It cannot prevent a dishonest artist from producing more physical copies than declared (it makes the violation legally demonstrable, not impossible)
- It cannot replace institutional reputation — it replaces the documentary vacuum
- Legal interpretations have not been validated by a jurist
- The pilot case (March 2026) lacked Layer 5 (PEC) — documented transparently in the paper

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Contacts

This is a working proposal. Critical contributions welcome.

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If you are a jurist, cryptographer, archivist, artist, or academic with something to add — write.